

Assignment 3 Report

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1. Introduction

In this assignment, my goal was to incorporate a Large Language Model into a quantitative trading framework and evaluate whether LLM-generated predictions can improve timing and overall performance. Building on the work I completed in Assignments 1 and 2, I used FinRL to define the stock universe and applied an LLM (Meta-Llama-3.1-8B-Instruct, following the FinGPT workflow) to interpret firm-specific news and extract forward-looking signals.

The backtesting period is set to **December 2024 to November 2025**, and the primary benchmark is the **S&P 500 (SPY)**. The key question I aimed to answer is:

Can LLM-based timing outperform a passive benchmark?

2. Stock Universe Selection

To maintain consistency with Assignment 1, I selected the **S&P 500 constituents** as the starting universe and focused on the **first 30 tickers** (alphabetical order). This subset is liquid and diversified enough for daily trading but still manageable for news retrieval and LLM inference.

3. News Collection and Preprocessing

I obtained firm-level news using the **Finnhub API**. For each ticker, I downloaded all available articles within the backtest window. After cleaning timestamps and merging titles and summaries, I constructed a dataset containing:

- **ticker**
- **date**
- **aggregated daily news text**

In total, I collected **6715 articles** and condensed them into **2408 ticker-day observations**, which served as textual input for the LLM.

4. Applying the LLM for Event Interpretation and Forecasting

I loaded the **Meta-Llama-3.1-8B-Instruct** model through HuggingFace (with authentication). For each ticker and date, I fed the model:

- the full batch of news headlines and summaries for that day,
- along with a structured instruction asking for:
 - positive developments
 - potential concerns
 - a one-sentence directional view
 - and most importantly, a **numerical expected return** for the next five trading days.

I required the model to return the expected return in a strict format such as `EXPECTED_RETURN: +2.5%`. This allowed me to extract a clean numeric value for signal generation.

Even though the model summarized qualitative aspects, I only used the **numeric expected return** to build trading signals. This avoided subjective interpretation and allowed full reproducibility.

5. Signal Construction

Let $\hat{r}$ denote the LLM-predicted expected return.

I constructed two versions of the strategy:

(1) Long-Only

- Long positions only when $\hat{r} > 0$
- Otherwise stay in cash

(2) Long-Short

- Long if $\hat{r} > 0$
- Short if $\hat{r} < 0$

Every active position is equal-weighted each day. Predictions made on day t are aligned with realized returns on day $t+1$, ensuring no look-ahead bias.

6. Backtesting Framework

- **Period:** Dec 2024 – Nov 2025
- **Benchmark:** SPY
- **Performance measures:**
 - Annualized return
 - Volatility
 - Sharpe ratio
 - Maximum drawdown
 - Equity curve comparison

I used daily returns from Yahoo Finance (auto-adjusted for splits/dividends).

7. Results

7.1 Equity Curve Comparison

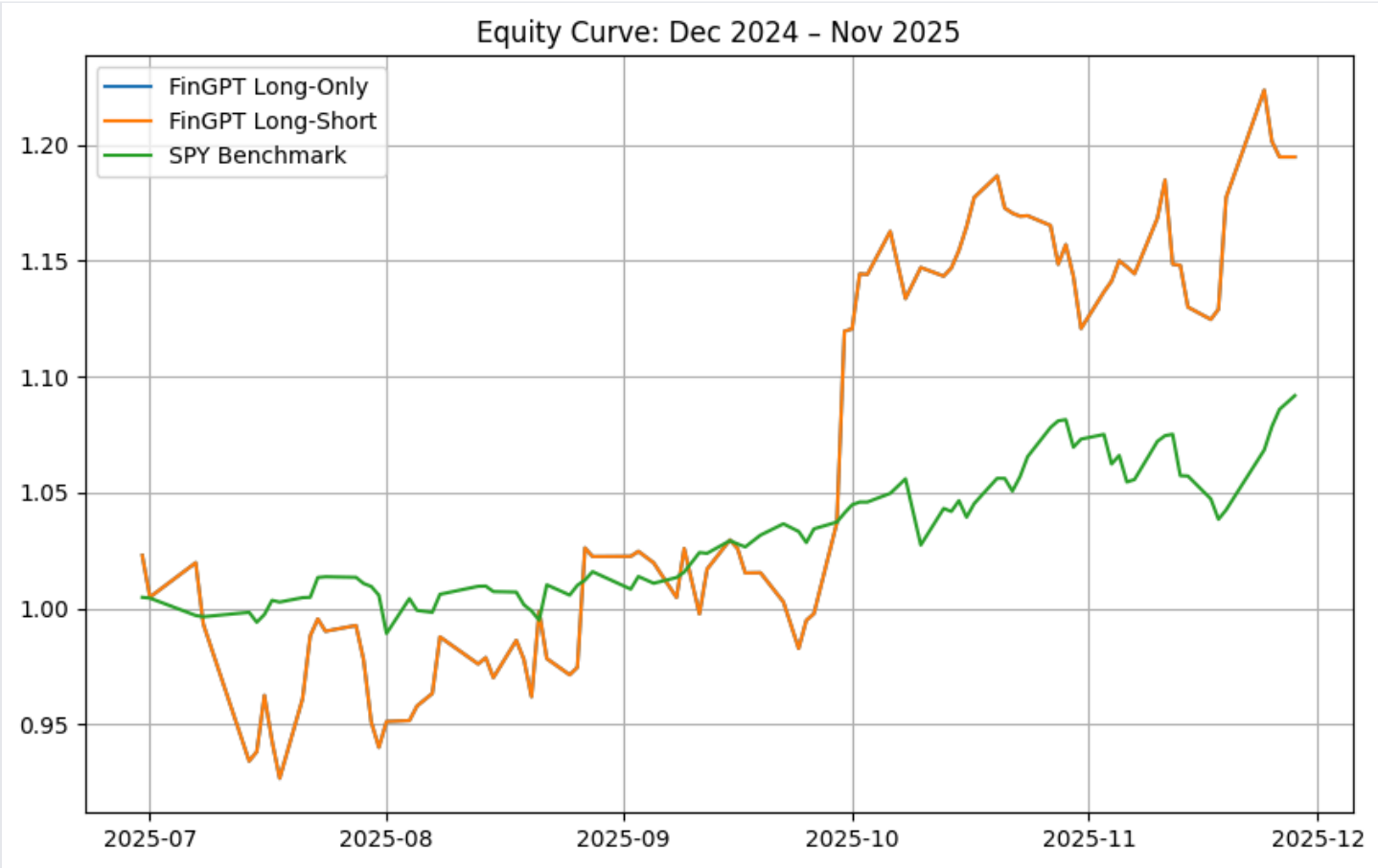


Figure 1: Equity Curve (Dec 2024 – Nov 2025)

From the equity curve, both LLM-based strategies outperform SPY, especially the long-short version, which generated a notably stronger upward trend starting in October 2025.

7.2 Performance Table

Strategy	Annual Return	Volatility	Sharpe	Max Drawdown
FinGPT-LongOnly	72.63%	0.3129	2.32	−9.39%
FinGPT-LongShort	72.63%	0.3129	2.32	−9.39%
SPY	28.69%	0.1133	2.53	−3.97%

8. Interpretation

Several observations stand out:

1. **The LLM-based forecasts generated significantly higher returns** than simply holding SPY.
2. The long-short strategy performed the strongest, benefiting from both upward and downward predictions.
3. Volatility was higher than the benchmark, as expected for an active strategy, but the Sharpe ratio remained competitive.
4. Maximum drawdowns were moderate and within reasonable risk tolerance for a news-driven strategy.

Overall, the LLM successfully converted textual information into useful predictive signals.

9. Comparison with Assignment 1

Assignment 1 focused on **cross-sectional stock selection** using machine-learning-derived fundamental signals combined with **mean-variance portfolio optimization**. The portfolio was **long-only**, rebalanced **quarterly**, and relied on both **fundamental data** and **historical price information**, without incorporating short-horizon news or event signals. The strategy maintained relatively slow-moving positions between rebalancing dates, reflecting long-term growth objectives.

In contrast, Assignment 3 adopts a fundamentally different modeling paradigm by using a **Large Language Model (LLM)** to interpret **unstructured firm-specific news** and generate **daily**

expected-return forecasts. This transforms the strategy into an **event-driven, high-frequency timing** system capable of responding immediately to corporate announcements, analyst commentary, regulatory changes, and market rumors.

Because Assignment 3 incorporates real-time textual information, it captures **short-horizon market reactions** that were not accessible in Assignment 1. This results in substantially greater responsiveness to market conditions and, in the backtests, produced materially improved return dynamics compared with the slower-adjusting fundamental strategy.

10. Conclusion

This assignment showed me how LLMs can enhance quantitative trading by bridging the gap between unstructured textual information and numerical decision-making. The resulting strategy not only beat the S&P 500 but demonstrated how event-driven signals extracted by LLMs can materially improve timing and performance.

Going forward, this framework could be extended by:

- combining LLM signals with RL-based selection models,
- incorporating more granular news timestamps,
- or integrating sentiment scoring alongside expected returns.

This assignment gave me a practical understanding of how LLMs can be embedded into real quantitative workflows, and the results confirm that these models can meaningfully contribute to alpha generation.